MILITARY SPECIFICATION

SALT PELLETS
(FOR USE IN AMMUNITION)

1. SCOPE

1.1 This specification covers a type of salt pellets prepared with lithium chloride and sodium ethylenediamine tetraacetic acid (Na_4EDTA) for use in Aerial Mines.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-G-155 - Graphite (For Use in Ammunition)
MIL-S-14752 - Lithium Chloride
MIL-S-14753 - Sodium Ethylenediamine Tetraacetic Acid (Na_4EDTA)
MIL-I-45607 - Inspection Equipment, Supply and Maintenance of

STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes (ABC-STD-105)
MIL-STD-109 - Quality Assurance Terms and Definitions
MIL-STD-129 - Marking for Shipment and Storage
MIL-STD-1168 - Lot Numbering of Ammunition
MIL-STD-1235 - Single and Multilevel Continuous Sampling Procedures and Tables for Inspection by Attributes

FSC: 1345

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DRAWINGS

U.S. ARMY MUNITIONS COMMAND

C9229382 - Pellet, Salt

PUBLICATIONS

ET - 9229382 - Equipment Tabulation

(Copies of specifications, standards, drawings and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the Contracting Officer.)

3. REQUIREMENTS

3.1 Form.-The pellets shall conform with all the requirements of drawing 9229382 and the specifications referenced thereon.

3.2 Chemical composition.-The pellets shall be homogeneous. (NOTE OF CAUTION: The Sodium EDTA (MIL-S-14753 (MU)) as received normally contains an excessive amount of moisture and therefore should be dried at 100 degrees C. for a suitable length of time prior to blending with the lithium chloride (MIL-L-14752 (MU)) and graphite (MIL-G-155). Failure to do so may result in production of salt pellets which do not meet the moisture requirement of drawing 9229382. In addition, the salt pellets shall have a minimum chelation value of 63 when tested as specified in 4.3.5.

3.3 Density.-The density of the pellets shall conform with the requirements of drawing 9229382 when tested as specified in 4.3.6.

3.4 Moisture.-The moisture content shall conform with the requirement of dwg. 9229382 when tested as specified in 4.3.1.

3.5 Workmanship.-The pellets shall be firmly compacted, free of soft spots and powdery areas, and shall have no more than trivial cracks or chips.

4. QUALITY ASSURANCE PROVISIONS

4.1 General quality assurance provisions.-Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform

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any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements. Reference shall be made to Standard MIL-STD-109 in order to define the terms used herein.

4.1.1 Contractor quality assurance system.- If the contractor desires to utilize a quality assurance system which is at variance with the quality assurance provisions of 4.2 and 4.3 and other documents referenced herein, he shall submit a written description of the system to the contracting officer for approval prior to initiation of production. It shall include a description covering controls for lot formation and identification, sampling procedures, methods of inspection (measuring and testing equipment), and provisions for control and disposition of non-conforming material. The written description will be considered acceptable when, as a minimum, it provides the quality assurance provisions required by the provisions of 4.2 and 4.3 and other documents referenced herein. The contractor shall not be restricted to the inspection station or the method of inspection listed in this specification provided that an equivalent control is included in the approved quality assurance procedure. In cases of dispute as to whether certain procedures of the contractor's system provide equal assurance, the comparable procedures of this specification shall apply. The contractor shall notify the Government of, and obtain approval for, any changes to the written procedure which affect the degree of assurance required by this specification or other documents referenced herein.

4.1.2 Submission of product.- At the time the completed lot of product is submitted to the Government for acceptance, the contractor shall supply the following information accompanied by a certificate which attests that the information provided is correct and applicable to the product being submitted:

a. A statement that the lot complies with all requirements and provisions of the approved current written description of the material.

b. Specification number and date, together with an identification and date of changes.

c. Certificates of analysis on all material used directly by the contractor when such material is controlled by Government specification shall be made available upon request by the contracting officer.

d. Quantity of product in the lot.

e. Date submitted.

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The certificate shall be signed by a responsible agent of the certifying organization. The initial certificate submitted shall be substantiated by evidence of the agent's authority to bind his principal. Substantiation of the agent's authority will not be required with subsequent certificates unless, during the course of the contract, this authority is vested in another agent of the certifying organization.

4.1.3 First article inspection

4.1.3.1 Submission.- The contractor shall submit a first article quantity as designated by the Contracting Officer for evaluation in accordance with the provisions of 4.1.3.2 (See 6.1). The first article sample shall consist of one thousand (1000) salt pellets. All salt pellets submitted shall have been produced by the contractor (or furnished by a supplier) using the same production processes, procedures, and equipment as will be used in fulfilling the contract. All materials, including packaging and packing, shall be obtained from the same sources of supply as will be used in regular production. The sample shall be accompanied by certificates of conformance. A first article quantity, or portion thereof, as directed by the Contracting Officer, shall also be submitted whenever there is a lapse in production for a period in excess of 90 days, or whenever a change occurs in manufacturing process, material used, drawing, specification or source of supply as to significantly affect product uniformity as determined by the Government. Prior to submission, the contractor shall inspect the sample to the degree necessary to assure that it conforms to the requirements of the contract and submit a record of this inspection with the sample. A sample containing known defects will not be submitted unless specifically authorized by the Contracting Officer.

4.1.3.2 Inspections to be performed.- The sample will be subjected by the Government to any or all of the examinations or tests specified in 4.2 and 4.3 of this specification and any or all requirements of the applicable drawings.

4.1.3.3 Rejection.- If any salt pellet fails to comply with any of the applicable requirements, the first article quantity shall be rejected. The Government reserves the right to terminate its inspection upon any failure of a sample to comply with any of the stated requirements. In the event of rejection, the Government

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reserves the right to require the contractor to take corrective action and submit a new first article quantity or portion thereof. Until a first article quantity is accepted, the contractor is in no way authorized by the Government to resume regular production unless otherwise directed by the Contracting Officer.

4.2 Inspection provisions

4.2.1 Lot formation.- A lot shall consist of salt pellets produced from the same batch of composition mix (maximum 450 pounds) by one manufacturer, in accordance with the same specification or same specification revision, under one continuous set of operating conditions. If the blend size is greater than 450 pounds, the blend should be divided into lots of 450 pounds or less. Each batch of salt pellets shall consist of that quantity of materials comprising the final composition which has been subjected to the same chemical or physical mixing process intended to make the final product homogeneous. The product shall be submitted for inspection in accordance with Standard MIL-STD-105 or Standard MIL-STD-1235 as applicable. The criteria and procedures for the assignment of lot numbers shall be in accordance with Standard MIL-STD-1168.

4.2.2 Examination.- Sampling plans and procedures for the following classifications of defects shall be in accordance with MIL-STD-105 (ABC-STD-105), except that inspection for Critical defects shall be 100 percent. Contractor's sampling plans, if used, shall be approved by the Government and shall provide, as a minimum, the protection afforded the Government by the sampling plans in MIL-STD-105. Continuous sampling plans in accordance with MIL-STD-1235 may be used if approved by the procuring activity. Also, at the option of the procuring activity, AQL's and sampling plans may be applied to the individual characteristics listed, using an AQL of 0.40 percent for each Major defect and an AQL of 0.65 percent for each Minor defect except where 100 percent inspection is specified. Equipment necessary for the performance of the inspections listed shall be in accordance with 4.2.4.

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4.2.2.1 Pellets (see dwg. 9229382)

Categories	Defects	Method of Inspection	Code No. (See 6.2)
Critical: None defined			
Major:	AQL 0.65 percent		
101.	Weight	Balance	01001
102.	Height	Gage	01002
103.	Diameter	Gage	01003
Minor:	AQL 0.65 percent		
201.	Evidence of poor workmanship (see 3.3)	Visual	01004

4.2.3 Testing

4.2.3.1 Composition & Moisture (Including Chelation Value).- Three (3) samples consisting of ten (10) pellets each shall be selected from each lot of pellets (maximum 450 pounds). The first set of ten shall be withdrawn from the first 46 pellets produced. The second set of ten pellets shall be withdrawn when approximately half of the lot has been produced, and the third set of ten pellets shall be withdrawn from the last 46 pellets produced. The samples selected for this test shall not be reworked, dried nor otherwise altered after the pelletizing operation. Each set of ten shall be pulverized, mixed and tested separately for conformance to 3.2, in accordance with the procedures given in 4.3.1 through 4.3.4. If any of the three (3) samples fails to comply with any of the requirements specified, the lot shall be rejected.

4.2.3.2 Density.-A random sample of 5 individual pellets shall be selected from the sample used in 4.2.2.1. If any of these pellets fails to meet the requirements, the lot shall be rejected.

4.2.4 Inspection equipment.-Equipment Tabulation Number ET-9229382 identifies the inspection equipment required to perform the examinations and tests prescribed in this section. The contractor shall design inspection equipment in accordance with the instructions in 6.3.

4.2.4.1 Government rights to documentation.-Inspection equipment drawings and lists provided and revised in accordance with the requirements of the ET may be used by DOD activities for design, procurement, manufacture, testing, evaluation, production

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and receiving inspection, overhaul, shipping, storage, identification of stock, ordering and storage of replacement parts, inspection of items at overhaul, general maintenance of equipment, construction, survey and whenever inspection equipment drawings are needed.

4.2.4.2 Supply and maintenance.- Supply and maintenance of the equipment listed on the ET shall be in accordance with MIL-I-45607.

4.2.4.3 Government use of contractor's inspection equipment.- The contractor shall make available to the Government all inspection equipment necessary for determining conformance with contract requirements. Personnel for operating the equipment and verification of its accuracy shall be supplied by the contractor for the performance of examination or test by the Government.

4.3 Test methods and procedures

4.3.1 Moisture - Major defect - (Defect Code 02001).- Transfer an accurately weighed portion of about two (2) grams of the sample to a tared weighed dish with a cover. Dry the dish and contents with cover removed for three (3) hours in an oven at 100 plus or minus 5 degrees C. Remove sample from oven and immediately cover the dish, cool in a desiccator and weigh. Determine the loss in weight and calculate the percentage of moisture as follows:

$$\text{Moisture, } \% = \frac{A\ (100)}{B}$$

where:

A = loss in weight, grams.
B = weight of sample, grams.

4.3.2 Graphite - Major defect - (Defect Code 03001).- Transfer an accurately weighed portion of approximately one (1) gram of the sample to a tared medium porosity glass filtering crucible attached to a clean suction flask apparatus. Extract the contents of the crucible with ten (10) 20 ml. portions of hot distilled water, using suction to aid filtering. Retain the filtrate for the determination given in 4.3.3. Dry the crucible and contents to constant weight in an oven maintained at 100 plus or minus 5 degrees C. Cool crucible and contents in a desiccator and weigh. Calculate the percentage of graphite as follows:

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$$\text{Graphite} = \frac{A\ (100)}{B}$$

where:

A = weight of residue in crucible, grams.
B = weight of sample, grams (moisture free).

4.3.3 Lithium chloride - Major defect - (Defect Code 04001).- Quantitatively transfer the filtrate obtained in 4.3.2 to a 500 ml. volumetric flask and make up to volume. Transfer a 100 ml. aliquot from the volumetric flask to a 500 ml. flask. Add 50 ml. of 3N-nitric acid plus exactly 40 ml. of 0.1N standardized silver nitrate and back titrate with 0.1N standardized potassium thiocyanate to a reddish-brown color using ferric ammonium sulfate indicator. Calculate the percentage of lithium chloride as follows:

$$\text{Lithium chloride, } \% = \frac{(4.24)\ (N_1 x V_1) - (N_2 x V_2)}{W}$$

where:

W = weight (corrected for moisture) of sample in aliquot used, grams.
N_1 = normality of silver nitrate solution.
V_1 = volume of silver nitrate, ml.
N_2 = normality of potassium thiocyanate solution.
V_2 = volume of potassium thiocyanate used.

4.3.4 EDTA - Major defect - (Defect Code 05001).- Calculate the percentage of EDTA by difference:

EDTA, % = 100% - (B plus C)

where:

B = percent graphite.
C = percent lithium chloride.

4.3.5 Chelation value - Major defect - (Defect Code 06001)

4.3.5.1 Reagents:

Ammonium Oxalate: 0.66% - dissolve 6.6g of ACS grade $(NH_4)_2C_2O_4.H_2O$ in a liter of hot distilled water.

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Sodium Hydroxide: 50% solution: dissolve 250g of reagent grade sodium hydroxide with stirring in 250 ml. of distilled water.

Calcium Chloride: 0.5M - add 50.05g of reagent grade calcium carbonate to 300 ml. of distilled water. Add slowly (use caution) 86 ml. of concentrated reagent grade hydrochloric acid. Heat to boiling. Cool and dilute to exactly one liter.

4.3.5.2 Transfer an accurately weighed portion of approximately 3 grams (NOTE: A pellet weighed accurately may be used as an alternate) of the sample to a 200 ml. beaker. Add 150 ml. of hot distilled water and stir until the soluble portion of the pellets are completely dissolved. Filter and wash the resultant solution through a number 40 Whatman filter paper catching the filtrate in a 300 ml. Erlenmeyer flask. Add 5 ml. of the 50% sodium hydroxide solution followed by 85 ml. of ammonium oxalate to the Erlenmeyer flask. Titrate the contents of the flask with the 0.5M calcium chloride to obtain a permanent but faintly turbid end point. Record this titration value and calculate as follows:

$$\text{Chelation Value} = \frac{\text{ml titer X50}}{\text{grams of sample (corrected for moisture)}}$$

4.3.6 Density.-The density of each pellet shall be determined by using the following calculation:

$$\text{Density} = \frac{\text{Weight, gms}}{(\text{height, inches})\ (\text{Diameter, inches})^2\ (12.86)}$$

5. PREPARATION FOR DELIVERY

5.1 Packaging

5.1.1 Level A.-The pellets shall be packaged in metal containers of the bolted ring cover style, conforming to Specification 37A of Interstate Commerce Regulations (Title 49 of the Code of Federal Regulations) except that bottom seams shall be compound lined and covers shall be fitted with a suitable tubular or flowed-in gasket securely affixed so that it will remain in position when covers are removed.

5.1.2 Level C.-The pellets shall be packaged in commercial containers which comply with the regulations applicable to the mode of transportation and which will afford the pellets protection against damage during shipment for the supply source to the first

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receiving activity. The container must provide protection against entrance of moisture which will be absorbed by the pellets.

5.2 Marking.- Marking shall be in accordance with MIL-STD-129.

6. NOTES

6.1 Ordering data.- Procurement documents shall specify the title, number and date of this specification, and provisions for submission of first article samples.

6.2 Inspection code numbers.- The five-digit code numbers assigned to the inspection herein are to facilitate future data collection and analysis by the Government.

6.3 Inspection equipment.- The contractor shall design inspection equipment as required by the referenced Equipment Lists (EL) in accordance with the instructions of paragraphs 6.3.1 through 6.3.6.

6.3.1 Equipment Lists.- Equipment lists indicate the availability of inspection equipment designs by showing in the "Part number" column of the list of inspection equipment (Form SMUPA 1010) the numbers of drawings or Federal Stock numbers of existing equipment designs, or codes as indicated in paragraph 6.3.2. Design action required of the contractor is described in paragraphs 6.3.3 and 6.3.4. The contractor will be required to prepare detailed drawings in accordance with 6.3.4 for all the equipment coded as "Contractor Design" in the "Part number" column. These contractor designs must be approved by the Government prior to fabrication or procuring of the equipment. Designs shall be submitted for approval as specified in 6.3.6. Action required by the contractor with respect to the use of commercial equipment is detailed in 6.3.5.

6.3.2 Equipment list codes.- The inspection equipment as defined in 6.3.3, 6.3.4 and 6.3.5 will be designated in the EL by the following codes:

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CD - Contractor design on controlled contractor format and/or commercial equipment.
MU - Army design, mandatory for use.

6.3.3 Army designs.- Army designs are reflected on detailed drawings which completely depict all the information necessary for the fabrication of the item of inspection equipment. The contractor need provide no design when an Army design is listed for an item of inspection equipment. Army designs fall into two basic classifications; mandatory (designated "MU") and non-mandatory (indicated by drawing or Federal Stock Number). When an equipment list references mandatory Army designs, the contractor shall comply with, and use these designs accordingly. The contractor may, however, in connection with non-mandatory designs, and with the approval of the Government, design alternate inspection equipment to facilitate his operations. Such contractor-prepared designs (see 6.3.4) or commercial equipment selections (see 6.3.5) must be approved by the Government prior to fabrication or procuring of the equipment. Designs shall be submitted for approval as specified in 6.3.6.

6.3.4 Contractor designs.- Contractor designs are designs of inspection equipment for which the Government has assigned design responsibility to the contractor. Contractor designs shall be supported by detailed drawings which depict all information necessary to completely fabricate, calibrate and operate an item of inspection equipment. This requires that the necessary views, dimensions, materials, finishes, notes and operating and calibration instructions be properly depicted in accordance with approved practices to the extent that further calculation or clarification will not be required. Unless otherwise specified, contractor designs may be developed on the format the contractor normally employs in his equipment design procedure provided such format reflects the detail and information specified above, subject to the following controls: All submitted contractor designs shall conform to Specification MIL-D-1000, Category E, Form 2. Legibility and reproducibility shall permit conventional making of clearly understandable, high-contrast reproductions. Contractors shall submit one copy of final designs as a rolled set. Designs shall be submitted for approval as specified in 6.3.6.

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6.3.5 Commercial equipment.- Commercial equipment is inspection equipment that has universal application for a specific function. It is comprised of items commonly used of commercial inspection equipment, but a list of such equipment must be approved by the Government. When commercial inspection equipment requires the use of special fixturing, the design of this fixturing shall be considered as part of the total design, and 6.3.4 shall apply. In the latter case, only those portions of the commercial equipment need be detailed in the designs as are necessary to completely fabricate, calibrate, and operate the total item of inspection equipment. Lists and fixturing designs shall be submitted for approval as specified in 6.3.6.

6.3.6 Submission of contractor designs.-All submitted designs shall reference the applicable five-digit code number contained in Section 4 of this specification. Unless otherwise specified on the EL, all designs of equipment for inspection of defects classified as critical and major shall be submitted for approval to the Commanding Officer, Picatinny Arsenal, ATTN: SMUPA-ND8. All other designs of inspection equipment shall be approved by the inspection element of the agency administering the contract. Submission shall be as directed by the contracting officer. Partial submission of inspection equipment designs is permissible and encouraged. However, the Arsenal completion date for design review will be based on the date of the final submission of designs. Picatinny Arsenal design review will be accomplished, normally, within one month after receipt.

6.4 Intended Use.-The material covered in this specification is intended for use in Aerial Mines.

Custodian:
Army-MU

Preparing activity:
Army-MU

Project Number: 1345-A-237

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